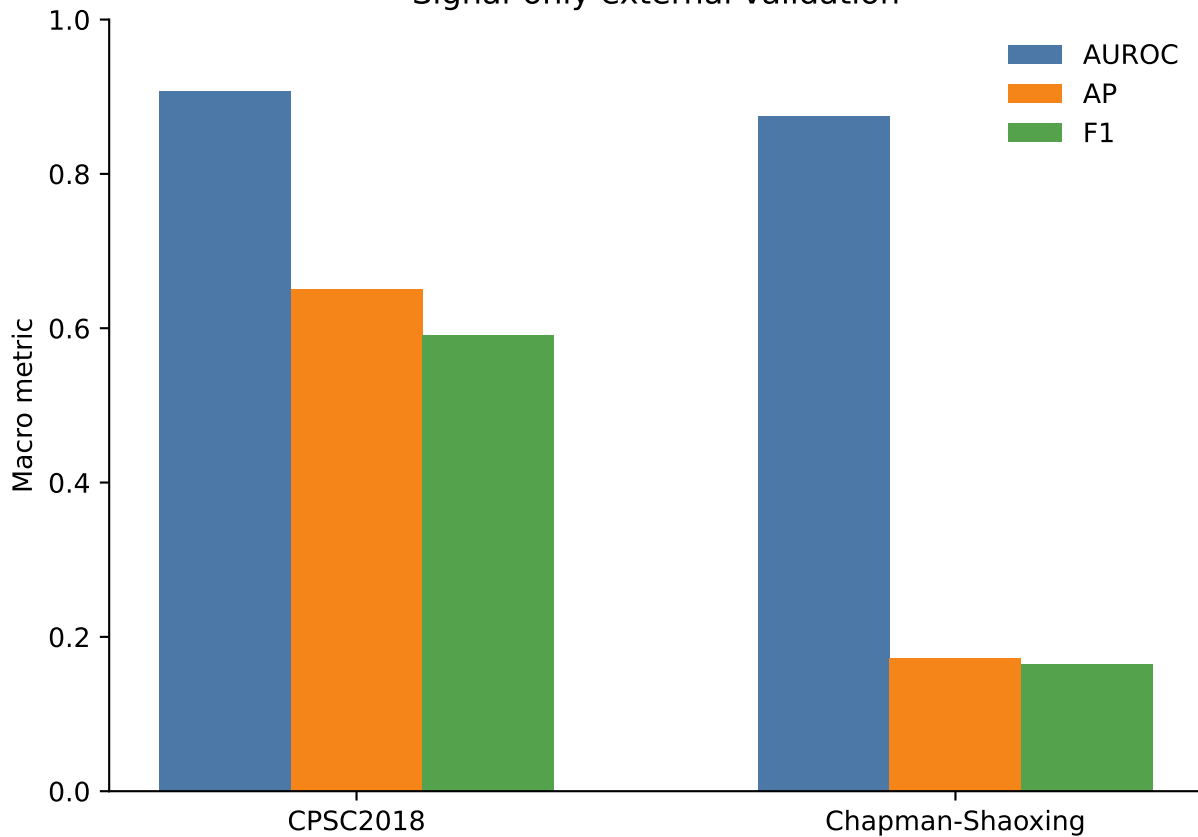


Signal-only external validation



External evaluation is signal-only. No external threshold tuning. Chapman AP/F1 reflect low prevalence, label mapping, and threshold transfer.